

MATHEMATICAL PRELIMINARIES

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WHY STUDY BANACH'S THEOREM?

- Last time: McCall's job search model—solved using **value function iteration**.
- **Key questions** we didn't fully answer:
 - Does a solution exist?
 - Is the solution unique?
 - Why does the iterative algorithm converge?
 - How fast does it converge?
- Today: **Banach's Contraction Mapping Theorem** answers all these questions.
- **Bonus**: It applies far beyond economics—ODEs, optimization, machine learning, image processing...
- This theorem is the **mathematical foundation** of dynamic programming.

MATHEMATICAL FOUNDATIONS

FIXED POINTS AND SELF-MAPS

- Before solving any problem it is useful to know if there is a solution at all.
- We will use a powerful theorem that will help us answer this question.
- First we introduce a concept of a **fixed point**.
- Let U be any nonempty set. We call T a self-map on U if $T : U \rightarrow U$.
- For a self-map T on U , we say that a point $u^* \in U$ is a **fixed point** of T if $Tu^* = u^*$.

FIXED POINT

- Some examples of fixed points:
 - $U = \mathbb{R}$, $T(u) = 2u + 3$. Then $u^* = -3$ is a fixed point of T .
 - $U = [0, 1]$, $T(u) = u$. Then every $u \in U$ is a fixed point of T .
 - $U = \mathbb{R}$, $T(u) = u + 1$. There is no fixed point of T .
 - $U = \mathbb{R}$, $T(u) = u^2$. Then $u^* = 0$ and $u^* = 1$ are both fixed points.
- **Global stability:** a self-map T on U is **globally stable** on U if T has a unique fixed point u^* in U and $T^k u \rightarrow u^*$ for all $u \in U$.

METRIC SPACE

- A metric space is a set U , together with a metric (distance function) $\rho, \rho : U \times U \rightarrow \mathbb{R}$, such that for all $u, v, w \in U$:
 - $\rho(u, v) \geq 0$ (nonnegativity), with equality if and only if $u = v$.
 - $\rho(u, v) = \rho(v, u)$ (symmetry).
 - $\rho(u, v) \leq \rho(u, w) + \rho(w, v)$ (triangle inequality).

METRIC SPACE

- **Convergence**: a sequence $\{x_n\}_{n=0}^{\infty}$ in U **converges** to $x \in U$, if for each $\epsilon > 0$, there exists N_{ϵ} such that $\rho(x_n, x) < \epsilon, \forall n \geq N_{\epsilon}$.
- **Cauchy sequence**: A sequence $\{x_n\}_{n=0}^{\infty}$ in U is **Cauchy** if for each $\epsilon > 0$, there exists N_{ϵ} such that $\rho(x_n, x_m) < \epsilon, \forall n, m \geq N_{\epsilon}$.
- **Complete metric space**: A metric space (U, ρ) is **complete** if every **Cauchy sequence** in U **converges** to an element in U .
- Example: $\mathbb{R}$ with $\rho(u, v) = |u - v|$ is a complete metric space.

WHY COMPLETENESS MATTERS FOR COMPUTATION

- **Key insight:** In iterative algorithms, we generate a sequence $u_0, u_1, u_2, \dots$ by repeatedly applying an operator.
- If consecutive terms get closer together ($\rho(u_{k+1}, u_k) \rightarrow 0$), the sequence is **Cauchy**.
- **Computational problem:** We can check if terms are getting closer, but we cannot directly verify convergence to a limit (we don't know the limit!).
- **Completeness to the rescue:** If the space is complete, being Cauchy *guarantees* convergence to some limit in the space.
- This means: If our algorithm produces a Cauchy sequence, we know it converges—even without knowing the limit ahead of time.

NORMED VECTOR SPACE

- A **normed vector space** is a vector space V (e.g., $\mathbb{R}^n$, space of functions) equipped with a norm $\|\cdot\| : V \rightarrow \mathbb{R}$ that measures "length" or "size".
- Properties for all $u, v \in V, \alpha \in \mathbb{R}$:
 - $\|u\| \geq 0$ (nonnegativity), with equality iff $u = 0$.
 - $\|\alpha u\| = |\alpha| \|u\|$ (homogeneity).
 - $\|u + v\| \leq \|u\| + \|v\|$ (triangle inequality).
- **Key link:** Every normed vector space is a metric space with distance $\rho(u, v) = \|u - v\|$.
- Examples on $\mathbb{R}^n$: $\|u\|_2 = \sqrt{\sum u_i^2}$ (Euclidean), $\|u\|_\infty = \max_i |u_i|$ (Supremum).

BANACH SPACE

- A **Banach space** is a normed vector space that is **complete** (every Cauchy sequence converges to a limit *within* the space).
- **Why it matters**: It ensures that if we construct a sequence of approximations that get closer together, they converge to a valid solution.
- **The space for DP**: Let $C(X)$ be the set of **bounded continuous** functions on X .
- With the supremum norm $\|f\|_{\infty} = \sup_{x \in X} |f(x)|$, $C(X)$ is a Banach space.
- This allows us to apply the Contraction Mapping Theorem to find value functions.

BANACH'S CONTRACTION MAPPING THEOREM

CONTRACTION

- Let (U, ρ) be a **metric space** and T a self-map on U . T is a **contraction mapping** (with modulus λ) if for some $\lambda \in [0, 1)$

$$\rho(Tu, Tv) \leq \lambda \rho(u, v), \quad \text{for all } u, v \in U.$$

- If T is a contraction on U , then T is uniformly continuous on U .
- **Note:** A contraction is more restrictive than uniform continuity. Not all continuous or uniformly continuous maps are contractions.

BANACH'S CONTRACTION MAPPING THEOREM

Theorem (Banach's contraction mapping theorem)

If (U, ρ) is a complete metric space and a self-map T is a contraction mapping with modulus λ , then:

- T has a **unique fixed point** u^* in U , and*
- for any $u_0 \in U$, $\rho(T^k u_0, u^*) \leq \lambda^k \rho(u_0, u^*)$ for all $k \in \mathbb{N}$.*

COMPUTATIONAL INTERPRETATION

- Banach's theorem provides the theoretical foundation for **fixed point iteration**:

$$u_{k+1} = Tu_k, \quad k = 0, 1, 2, \dots$$

- Algorithm guarantee**: Starting from any u_0 , the sequence converges to the unique fixed point u^* .
- Convergence rate**: The error decreases geometrically: $\rho(u_k, u^*) \leq \lambda^k \rho(u_0, u^*)$.
- Stopping criterion**: We can stop when $\rho(u_{k+1}, u_k)$ is small enough:

$$\rho(u_k, u^*) \leq \frac{\lambda}{1 - \lambda} \rho(u_k, u_{k-1}).$$

- This bound allows us to estimate how close we are to the solution without knowing u^* .

PROOF OF BANACH'S THEOREM

PROOF OUTLINE

- **Step 1:** Show that the sequence $\{u_k\}_{k=0}^{\infty}$ with $u_{k+1} = Tu_k$ is Cauchy.
- **Step 2:** Show that the limit u^* of this sequence is a fixed point.
- **Step 3:** Show that the fixed point is unique.
- **Step 4:** Prove the convergence rate bound.

PROOF: STEP 1 – CAUCHY SEQUENCE

- Define $u_{k+1} = Tu_k$ for $k = 0, 1, 2, \dots$
- By the contraction property:

$$\rho(u_2, u_1) = \rho(Tu_1, Tu_0) \leq \lambda \rho(u_1, u_0).$$

- By induction:

$$\rho(u_{k+1}, u_k) \leq \lambda^k \rho(u_1, u_0), \quad k = 0, 1, 2, \dots$$

PROOF: STEP 1 (CONTINUED)

- Using the triangle inequality, for $m > n$:

$$\begin{aligned}\rho(u_m, u_n) &\leq \rho(u_m, u_{m-1}) + \rho(u_{m-1}, u_{m-2}) + \cdots + \rho(u_{n+1}, u_n) \\ &\leq \left[\lambda^{m-1} + \lambda^{m-2} + \cdots + \lambda^n \right] \rho(u_1, u_0).\end{aligned}$$

- Factor out λ^n and use the geometric series formula:

$$\begin{aligned}\rho(u_m, u_n) &= \lambda^n \left[1 + \lambda + \cdots + \lambda^{m-n-1} \right] \rho(u_1, u_0) \\ &\leq \lambda^n \frac{1 - \lambda^{m-n}}{1 - \lambda} \rho(u_1, u_0) \leq \frac{\lambda^n}{1 - \lambda} \rho(u_1, u_0).\end{aligned}$$

- Since $\lambda < 1$, the right side $\rightarrow 0$ as $n \rightarrow \infty$, so $\{u_k\}_{k=0}^{\infty}$ is **Cauchy**.

PROOF: STEP 2 – FIXED POINT

- Since U is **complete**, the Cauchy sequence $\{u_k\}_{k=0}^{\infty}$ converges to some $u^* \in U$.
- We now show that u^* is a fixed point of T .
- For all n and all $u_0 \in U$:

$$\begin{aligned}\rho(Tu^*, u^*) &\leq \rho(Tu^*, T^n u_0) + \rho(T^n u_0, u^*) \\ &\leq \lambda \rho(u^*, T^{n-1} u_0) + \rho(T^n u_0, u^*).\end{aligned}$$

- Since $T^n u_0 \rightarrow u^*$, both terms $\rightarrow 0$ as $n \rightarrow \infty$.
- Thus $\rho(Tu^*, u^*) = 0$, which implies $Tu^* = u^*$.

PROOF: STEP 3 – UNIQUENESS

- Suppose there exists $\hat{u} \neq u^*$ such that $T\hat{u} = \hat{u}$.
- Then $\rho(\hat{u}, u^*) = \delta > 0$, and by the contraction property:

$$\delta = \rho(u^*, \hat{u}) = \rho(Tu^*, T\hat{u}) \leq \lambda \rho(u^*, \hat{u}) = \lambda \delta.$$

- This implies $\delta \leq \lambda \delta$, or equivalently $1 \leq \lambda$.
- But this **contradicts** $\lambda < 1$, so u^* is the **unique** fixed point.

PROOF: STEP 4 – CONVERGENCE RATE

- For any $k \geq 1$, using the contraction property:

$$\rho(T^k u_0, u^*) = \rho(T(T^{k-1} u_0), T u^*) \leq \lambda \rho(T^{k-1} u_0, u^*).$$

- By induction on k :

$$\rho(T^k u_0, u^*) \leq \lambda^k \rho(u_0, u^*).$$

- This shows **linear convergence** with rate λ .
- Alternative bound:

$$\rho(T^k u_0, u^*) \leq \frac{\lambda^k}{1 - \lambda} \rho(u_0, T u_0).$$

APPLICATIONS

AN EXAMPLE

- Let $U = \mathbb{R}$ with $\rho(u, v) = |u - v|$ and $T(u) = 0.5u + 3$.
- (U, ρ) is a **Banach space** (complete normed vector space).
- Check contraction property:

$$\rho(T(u), T(v)) = |0.5u + 3 - 0.5v - 3| = 0.5 |u - v| = 0.5\rho(u, v).$$

- So T is a **contraction** with modulus $\lambda = 0.5$.
- By **Banach's CMT**, there exists a **unique fixed point**. Solving $u^* = 0.5u^* + 3$ yields $u^* = 6$.
- For any u_0 : $|T^k u_0 - 6| \leq 0.5^k |u_0 - 6|$.

APPLICATION: DIFFERENTIAL EQUATIONS (1/2)

- The **Picard-Lindelöf Theorem** guarantees existence and uniqueness for ODEs.
- Consider the initial value problem (IVP):

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0.$$

- **Conditions:** Let D be a domain around (t_0, y_0) . If f is:
 1. Continuous in t and y , and
 2. Lipschitz continuous in y (uniformly in t): $|f(t, y_1) - f(t, y_2)| \leq L |y_1 - y_2|$.
- **Result:** There exists a **unique solution** $y(t)$ on some interval $[t_0, t_0 + \epsilon]$ for sufficiently small $\epsilon > 0$.

APPLICATION: DIFFERENTIAL EQUATIONS (2/2)

- **Key Idea:** Convert the differential equation to an integral equation:

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds.$$

- Define an operator T on the space of continuous functions $C([t_0, t_0 + \epsilon])$:

$$(Ty)(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds.$$

- A solution to the IVP is exactly a **fixed point** of T (i.e., $Ty = y$).
- **Why is T a contraction?** The Lipschitz condition ensures that for small enough ϵ , we have
$$\|Ty_1 - Ty_2\|_{\infty} \leq L\epsilon \|y_1 - y_2\|_{\infty} < \|y_1 - y_2\|_{\infty}.$$
- By Banach's CMT, a unique fixed point (solution) exists.

APPLICATION: SOLVING LINEAR SYSTEMS

- Consider solving $Ax = b$ where $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$.
- Rewrite as $x = (I - A)x + b$ and define the iteration:

$$x_{k+1} = (I - A)x_k + b.$$

- The operator $T(x) = (I - A)x + b$ is a contraction if $\|I - A\| < 1$ in some matrix norm.
- **Sufficient condition:** If the spectral radius $\rho(I - A) < 1$, i.e., all eigenvalues λ of A satisfy $|1 - \lambda| < 1$.
- **Geometric interpretation:** Eigenvalues of A must lie in the open disk of radius 1 centered at 1 (in the complex plane).
- **Result:** The iteration converges to $x^* = A^{-1}b$. Foundation of Jacobi and Gauss-Seidel methods.

APPLICATION: NEWTON'S METHOD

- To find a root of $f(x) = 0$ where $f : \mathbb{R} \rightarrow \mathbb{R}$, Newton's method iterates:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = g(x_k).$$

- Near a simple root x^* (where $f(x^*) = 0$ and $f'(x^*) \neq 0$), the operator g is a contraction.
- Why?** By Taylor's theorem, $g'(x^*) = \frac{f(x^*)f''(x^*)}{[f'(x^*)]^2} = 0$ (since $f(x^*) = 0$), so g is locally a contraction.
- Banach's CMT guarantees **linear convergence**: $|x_{k+1} - x^*| \leq \lambda |x_k - x^*|$ for some $\lambda < 1$.
- Beyond Banach**: Newton's method actually exhibits **quadratic convergence** near the root: $|x_{k+1} - x^*| \leq C|x_k - x^*|^2$ (from the special structure $g'(x^*) = 0$).
- This explains why Newton's method is so effective in practice.

APPLICATION: IMPLICIT FUNCTION THEOREM

- Consider an equation $F(x, y) = 0$ where $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$.
- **Question:** Can we solve for y as a function of x , i.e., find $y = \phi(x)$ such that $F(x, \phi(x)) = 0$?
- **Implicit Function Theorem:** If $F(x_0, y_0) = 0$ and $\frac{\partial F}{\partial y}(x_0, y_0)$ is invertible, then there exists a neighborhood of x_0 where $y = \phi(x)$ exists and is unique.
- **Constructive proof via Banach:** Define the operator:

$$T_x(y) = y - \left[\frac{\partial F}{\partial y}(x_0, y_0) \right]^{-1} F(x, y).$$

- For x close to x_0 , T_x is a contraction, and its fixed point gives $\phi(x)$.
- This provides not just existence, but a **constructive algorithm** to compute ϕ .

TOOLS FOR DYNAMIC PROGRAMMING

BLACKWELL'S SUFFICIENT CONDITIONS

- Verifying the contraction property directly can be tedious.
- **Blackwell's sufficient conditions** provide a convenient alternative for operators on the space of bounded functions $B(X)$ with the sup norm.
- An operator $T : B(X) \rightarrow B(X)$ is a contraction with modulus β if:
 1. **Monotonicity**: For any $f, g \in B(X)$, $f(x) \leq g(x)$ for all x implies $Tf(x) \leq Tg(x)$ for all x .
 2. **Discounting**: There exists $\beta \in (0, 1)$ such that for any $f \in B(X)$ and constant $c \geq 0$,

$$T(f + c)(x) \leq Tf(x) + \beta c, \quad \text{for all } x.$$

WHY BLACKWELL'S CONDITIONS IMPLY CONTRACTION

- We need to show: $\|Tf - Tg\|_{\infty} \leq \beta \|f - g\|_{\infty}$ for all f, g .
- Let $c = \|f - g\|_{\infty}$. Then for all x : $-c \leq f(x) - g(x) \leq c$.
- This means: $g(x) \leq f(x) \leq g(x) + c$ for all x .
- By **monotonicity**: $Tg(x) \leq Tf(x)$ for all x .
- By **discounting**: $Tf(x) \leq T(g + c)(x) \leq Tg(x) + \beta c$ for all x .
- Combining: $Tf(x) - Tg(x) \leq \beta c$ for all x .
- By symmetry: $Tg(x) - Tf(x) \leq \beta c$ for all x .
- Therefore: $\|Tf - Tg\|_{\infty} \leq \beta \|f - g\|_{\infty}$.



THE TYPICAL WORKFLOW IN DYNAMIC PROGRAMMING

- Bellman equation for optimal value function:

$$v(x) = \max_{a \in (X)} \left\{ r(x, a) + \beta \int v(x') Q(dx' | x, a) \right\}.$$

- Bellman operator T on suitable function space:

$$(Tv)(x) = \max_{a \in (X)} \left\{ r(x, a) + \beta \int v(x') Q(dx' | x, a) \right\}.$$

- Step 1: Verify Blackwell's conditions for T .
- Step 2: Conclude T is a contraction with modulus β .
- Step 3: Apply Banach's CMT:
 - Unique fixed point v^* exists: $Tv^* = v^*$.
 - Value iteration $v_{k+1} = Tv_k$ converges to v^* .